

Exploratory Data Analysis (EDA) Report

Loan Default Prediction

Objective

The objective of this analysis is to understand the characteristics of borrowers, identify patterns associated with loan defaults, detect anomalies, and generate business insights that will guide feature engineering, machine learning model development, and business decision-making.

1. Dataset Overview

The dataset contains 32,416 loan records and 12 variables. The variables include borrower demographics, financial information, loan characteristics, credit history, and the target variable (loan_status), where 1 indicates a default and 0 indicates a non-default.

2. Data Quality Assessment

Duplicate records identified: 0.

Remaining missing values:

person_age	0
person_income	0
person_home_ownership	0
person_emp_length	0
loan_intent	0
loan_grade	0
loan_amnt	0
loan_int_rate	0
loan_status	0
loan_percent_income	0
cb_person_default_on_file	0
cb_person_cred_hist_length	0

3. Overall Loan Default Analysis

The overall loan default rate is 21.87%. This indicates that approximately 21.9% of borrowers in the dataset failed to repay their loans. This metric serves as the baseline against which predictive models will be evaluated.

4. Loan Purpose Analysis

- DEBTCONSOLIDATION: 28.68% default rate
- MEDICAL: 26.76% default rate
- HOMEIMPROVEMENT: 26.15% default rate
- PERSONAL: 19.90% default rate
- EDUCATION: 17.25% default rate

- VENTURE: 14.85% default rate

The highest-risk loan purpose is DEBTCONSOLIDATION, suggesting borrowers requesting loans for this purpose experience the greatest repayment difficulties.

5. Loan Grade Analysis

- Grade G: 98.44% default rate
- Grade F: 70.54% default rate
- Grade E: 64.49% default rate
- Grade D: 59.06% default rate
- Grade C: 20.75% default rate
- Grade B: 16.32% default rate
- Grade A: 9.96% default rate

Higher-risk grades generally correspond to higher default rates, confirming that the grading system effectively reflects borrower credit risk.

6. Home Ownership Analysis

- RENT: 31.61% default rate
- OTHER: 31.13% default rate
- MORTGAGE: 12.62% default rate
- OWN: 7.49% default rate

RENT borrowers exhibit the highest default rate, suggesting housing status influences financial stability and repayment behavior.

7. Previous Credit Default

Borrowers with previous defaults (Y) have a default rate of 37.87%, compared with 18.43% for borrowers without previous defaults. Previous repayment history is therefore a strong indicator of future loan performance.

8. Correlation Analysis

The strongest numerical relationships with loan default are:

- loan_percent_income: correlation = 0.380
- loan_int_rate: correlation = 0.320
- person_income: correlation = -0.145
- loan_amnt: correlation = 0.106
- person_emp_length: correlation = -0.083
- person_age: correlation = -0.023

9. Key Findings

- Loan default is influenced by multiple borrower and loan characteristics rather than a single factor.

- Borrowers with previous credit defaults are substantially more likely to default again.
- Certain loan purposes consistently exhibit higher default rates.
- Loan grade is strongly associated with repayment performance.
- Home ownership status appears to influence borrower risk.
- The identified patterns provide strong predictive signals for machine learning models.

10. Business Recommendations

- Introduce stricter approval criteria for high-risk loan purposes.
- Use previous default history as a key feature in credit scoring.
- Review underwriting policies for high-risk loan grades.
- Monitor applicants with high loan-to-income ratios more closely.
- Deploy the predictive model as a decision-support tool rather than replacing human review.